

Learned encoder/decoder inverse experiment plan

Decision and scope

The encoder should remain a learned representation model. This plan does **not** initialize it as an identity map, reserve physical-state channels, tie its weights to the decoder, or require a global mathematical inverse. The evidence does not justify those restrictions: in the completed one-degree, processor-bypassed factorial, the learned Perceiver encoder with a direct decoder reached MSE 0.025431, compared with 0.012083 for the direct encoder and decoder. Changing only the decoder increased MSE to 0.279303. The decoder is the primary failure; the existing encoder is a viable starting point.

The operational requirement is instead

```
z_0      = E_state,r(x_t)
q_m      = E_boundary,r(b_m), m = 1, ..., N
z_m      = P(z_{m-1}, q_m, g_r)
x_hat_s,m = D_s(z_m; g_r, g_s)
```

where r and s denote input and output grids and g contains grid geometry. At processor depth zero, same-grid reconstruction should satisfy $D_r(E_r(x)) \approx x$. For different grids, the target is the same physical field on the requested output grid, not an exact algebraic inverse. The decoder should also remain usable after any supported processor depth. This is a learned left-inverse condition on ocean states, not a demand that every latent vector or every encoder input (history and forcing included) be recoverable.

The selected decoder is:

```
native_channels = pointwise_decode(z_m)
prediction      = channel_masked_coordinate_resample(
                    native_channels, source_grid, output_grid
                )
```

The encoder and pointwise projection use ordinary learned initialization and train jointly as the zero-depth inverse. The zero-initialized local-attention correction remains implemented as a fallback, but the completed screen does not justify its cost and it is not part of the selected path. The implementation is:

- production resampling base: `src/samudra/models/modules/decoder.py`, `ResampleProjectionDecoder`;
- production hybrid: the same module's `ResampleAttentionResidualDecoder`;
- position-anchored correction: the same module's `LocalCoordinateAttentionCorrection`;
- zero-to-N model path and zero-depth decode: `src/samudra/models/samudra_multi.py`;
- processor geometry sidecar: `src/samudra/models/modules/augment_input.py`, `ProcessorGeometryConditioner`.

The former `use_fine_scale_queries` feature has been removed. That path could carry the original prognostic input around the processor, allowing the decoder to reconstruct without using $P^k(E(x))$. Static geometry and future forcing may condition the processor and decoder, but the prognostic state must pass through the latent route.

Implementation status

The development prerequisites are now present on this branch:

- identity diagnostics use disjoint deterministic train and held-out samples;
- `identity_eval_only: true` loads an explicit finetune checkpoint and evaluates every fixed held-out route once, without a redundant training-window pass, backward passes, or optimizer updates;
- paired reconstruction can read a destination resolution at the exact input timestamps, balance fixed samples across shape-distinct routes, and report both learned and deterministic-resampler errors in normalized and physical units;
- the diagnostic separately reports interpolation in physical space and directly in the source-normalized coordinate system, exposing cross-resolution errors caused by resolution-specific means and standard deviations;
- encoder geometry can be additive, absent, or supplied to each processor call as a zero-initialized position/scale sidecar;
- physical-coordinate resampling and the bounded zero-initialized local-attention correction are production decoder options;
- `SamudraMulti` exposes encode/process/decode, supports one shared processor at depths 0, 1, 2, 4, or any other non-negative integer when widths match, and has gradient tests across repeated applications;
- forecast training can opt into a same-grid zero-depth reconstruction auxiliary loss through `zero_depth_reconstruction_weight`; and
- a zero-depth encoder/decoder checkpoint can initialize a processor-present finetune with `finetune_allowed_missing_prefixes: ["processor.", "processor_geometry.", "boundary_encoder.", "processor_residual_scale"]`; every non-allowlisted missing key and every unexpected checkpoint key remains fatal, and EMA restarts for the composed model; and
- the synthetic probe now learns its encoder rather than copying target channels into the decoder input, uses fresh analytic coefficients and disjoint evaluation coefficients, and reports amplitude, bias, and high-wavenumber diagnostics.

The checked-in historical zero-depth controls are `model_learned_inverse_resample.yaml` and `model_learned_inverse_hybrid.yaml`. The selected state-only inverse and physical latent-autoregressive path are `model_learned_inverse_native_masked_projection.yaml` and `model_iterable_inverse_native_masked_projection.yaml`. The candidate set and defaults below remain revisable under the selection-logic rule.

The completed learned-encoder S0 screen supplies the first revision. At 2,000 updates and three seeds, physical-coordinate resampling plus projection beat both the hybrid and position-only attention on same-grid, downsampled, longitude-shifted, and upsampled routes. S1 therefore promotes the simple physical resampler. The hybrid remains implemented as a fallback but does not receive a parallel ocean sweep unless S1 spectra reveal an error the base cannot represent.

Questions to resolve

The experiments are designed to answer six questions separately.

1. Can a jointly learned encoder and resampling decoder generalize the zero-depth reconstruction map to held-out ocean states rather than merely fit a fixed set?
2. Do the encoder's additive position and scale embeddings help the processor, or do they make a shared pointwise decoder spend capacity removing location-dependent signals from the content representation?
3. Is position/scale information best added to content, omitted from the encoder output, or carried separately as immutable geometry?
4. Does a local position-anchored residual improve reconstruction and cross-resolution rendering beyond the resampling base without damaging the stable base route?
5. Should attention keys be normalized while values remain unnormalized so physical amplitude does not have to pass through LayerNorm?
6. Does a candidate retain zero-depth reconstruction while the processor is trained and evaluated at multiple iteration counts?

Required development

Implement these changes in order. Each step should land with focused unit tests and should be usable without enabling later steps.

D0. Make the reconstruction diagnostic measure generalization

Extend `src/samudra/identity.py` with independent deterministic training and held-out sample sets. The current diagnostic repeatedly trains and evaluates on the same fixed validation batches, which is useful for capacity tests but cannot distinguish a learned inverse from memorization.

Proposed configuration fields:

```
identity_train_samples: 32
identity_eval_samples: 32
identity_train_offset: 0
identity_eval_offset: 32
identity_input_source: 0
identity_output_source: 0
processor_iterations: 0
```

Log train and held-out metrics separately. Retain all existing channel, variable, depth, spectrum, high-wavenumber, and seam diagnostics. Add:

- prediction standard-deviation ratios and mean bias in target-standard-deviation units by channel;
- base MSE, final MSE, and correction-only MSE;
- `RMS(correction) / RMS(base)`;
- encoder, base-decoder, and correction gradient norms;
- parameter count, peak memory, samples/second, and optimizer updates.

Add an evaluation mode that loads one checkpoint and evaluates every supported same-resolution and cross-resolution source/target pair without further fitting.

D1. Make encoder geometry injection configurable

The current `PerceiverEncoder` adds learned position and scale embeddings directly to its content output. Add one encoder option with three modes:

```
encoder:
  geometry_mode: additive # additive | none | sidecar
```

- `additive` is the current behavior and remains the control.
- `none` removes the post-Perceiver position and scale additions. Intra-patch Fourier coordinates remain available to the Perceiver, so this does not make the encoder spatially unaware inside a patch.
- `sidecar` keeps content and geometry separate. The encoder returns learned content plus immutable source-grid coordinates/scale. The decoder uses the geometry for resampling and attention routing, and the processor receives it as conditioning at every iteration rather than as a one-time addition to content.

Do not silently concatenate geometry into the reconstructive tensor; that would confound geometry mode with latent width. Source coordinates now travel separately to the coordinate-aware decoder. At processor depth zero, `none` and `sidecar` are therefore the same model and must not be reported as independent experimental arms. Complete the distinction in D4 by conditioning every processor application on the sidecar, then compare `none` with `sidecar` only in processor-present runs.

Tests must establish:

- `additive` is bitwise-equivalent to the current default;
- `none` changes only the post-Perceiver additions;
- geometry tensors are deterministic, batch-independent, correctly periodic in longitude, and excluded from decoder content values;
- a change in output resolution changes routing geometry but does not mutate the learned source content.

D2. Replace shape-only resizing with physical-coordinate resampling

`ResampleProjectionDecoder` currently calls `F.interpolate` using only output shape. Introduce a coordinate resampler that consumes source and target latitude and longitude arrays. Its first implementation should be deterministic and have no learned parameters:

- periodic linear interpolation in longitude;
- latitude interpolation using actual cell-center coordinates;
- explicit masks for invalid/padded source cells;
- exact no-op when source and target coordinates match;
- float32 interpolation accumulation under bfloat16 model execution.

Keep the current `F.interpolate` implementation as a named control. Unit tests should cover identical grids, longitude wraparound, shifted longitude grids, nonuniform latitude, 1-to-1/2-degree upsampling, 1/2-to-1-degree downsampling, masks, and gradients.

D3. Promote the hybrid decoder from probe to production code

Add a production decoder containing the coordinate-resampling base and a local position-anchored attention correction. The attention implementation should use:

```
Q = projection(output coordinates and optional query features)
K = projection(LayerNorm(source content))
V = projection(source content)                # no LayerNorm on values
```

This isolates normalization to routing. The base path never passes through LayerNorm. Initialize only the correction's final output projection to zero. Add a configuration switch for the old normalized-value behavior as an ablation, not as the default.

The correction must use a bounded physical neighborhood rather than attend to an entire output window. Start with a 3-by-3 or 5-by-5 source-cell neighborhood, periodic longitude, a validity mask, and relative physical-coordinate features. Do not introduce a second learned latent bank.

Required tests:

- the hybrid is exactly equal to its base at initialization;
- both branches receive gradients after one optimizer update;
- raw-value and normalized-value modes have the expected data paths;
- same-grid, upsampled, downsampled, shifted, periodic, and masked cases work;
- changing output grid does not change parameter shapes;
- local attention memory scales linearly with output pixels times neighborhood size, not quadratically with global grid size.

D4. Expose encode, process, and decode and make processor iteration legal

Refactor `SamudraMulti.forward_once` into independently callable `encode`, `process`, and `decode` operations, while preserving the existing one-call behavior. The current encoder emits 128 channels and the configured UNet emits 380, so the same processor cannot simply be applied twice. Introduce an explicit canonical latent width and require the repeated processor core to map that width to itself.

The least disruptive first control is:

- learned Perceiver encoder output width equals the processor's first/output width;
- the UNet processor maps `C` -> `C`;
- the decoder always accepts `C`, including at iteration zero;
- one shared processor instance is applied zero through `N` times.

Start with `C=380`, matching the existing processor output. A later width sweep may change the whole processor consistently; do not place an iteration-dependent adapter between `P` applications. Feed sidecar geometry to every processor application in the same way.

Add equivalence tests for the existing one-processor path and tests for iterations `0`, `1`, `2`, and `4`, including checkpointing and gradients. Log reconstruction at iteration zero throughout forecast training so the encoder/decoder contract cannot regress unnoticed.

The processor contract is now settled: one application is one physical time step. For step m , separately encode exactly one aligned boundary state b_m , pass it to the shared processor, and keep the prognostic state latent:

```
z_0 = E_state(x_t)
z_m = P(z_{m-1}, E_boundary(b_m), geometry)
x_hat_{t+m} = D(z_m)
```

Depth N is supervised by x_{t+N} . The decoder is an output/supervision head; it is never fed back through the state encoder. Inference chunking must carry z_m , not the last decoded field. Backward checkpoint compatibility is explicitly out of scope: retrain the corrected state-only inverse rather than migrating the joint state-and-boundary encoder.

D5. Add losses without constraining the encoder architecture

Use the same learned encoder in every arm. For reconstruction-only runs optimize:

```
L_recon = normalized_MSE(D_r(E_r(x)), x)
```

For cross-resolution pairs optimize:

```
L_cross = normalized_MSE(D_s(E_r(x_r)), x_s)
```

For forecast training retain the existing forecast loss and add a zero-depth reconstruction batch or auxiliary term:

```
L = L_forecast + lambda_0 * L_recon
```

Sweep λ_0 only after a candidate passes the reconstruction funnel. Do not add a latent-cycle loss initially: $E(D(z)) \approx z$ would constrain unreachable latent directions and is stronger than the stated requirement.

Small-scale experiment funnel

Use successive halving rather than a full Cartesian sweep. Every comparison must retain the same fixed sample indices, output target, seed, optimizer-update count, precision, and mask.

S0. Synthetic coordinate and normalization screen

Extend `scripts/probe_perceiver_decoder.py` so its input first passes through a small learned encoder; do not copy target channels into latent channels. Train on fresh analytic fields and evaluate unseen coefficients.

Common settings:

Setting	Value
Source grids	8x8, 16x16
Output grids	same grid, 8->16, 16->8, half-cell longitude shift

Setting	Value
Channels	16 input/output, 32 learned latent
Train/eval batches	fresh coefficients; 256 unseen evaluation samples
Optimizer	Adam, no weight decay
Updates	500 screen, 2,000 finalists
Batch size	32
Learning rate	3e-4 , 1e-3 , 3e-3 on the base control; freeze the winner
Seeds	0 and 1 for screens; add 2 for finalists

Compare:

1. shape-only bilinear resampling plus projection;
2. physical-coordinate resampling plus projection;
3. position-anchored attention alone;
4. physical resampling plus zero-init attention correction.

For the hybrid, sweep in this order:

- attention values: {LayerNorm(value), raw value};
- neighborhood: {3x3, 5x5};
- position-bias strength: {1, 2, 4, 8, 16} using a 100-update screen, then train the best two for 500 updates;
- heads and head width: {1x64, 2x64, 4x32} only after choosing neighborhood and bias;
- correction output initialization: {exact zero, std=1e-3} as a confirmation that exact zero does not delay useful learning.

Promote raw values unless normalized values improve held-out error by at least 10% without amplitude bias. A candidate must preserve same-grid performance while improving at least one off-grid direction relative to physical interpolation.

S1. One-degree learned-inverse screen

Use the existing learned Perceiver encoder, processor depth zero, the one-degree dataset, 32 deterministic training samples, and 32 disjoint held-out samples.

Common settings:

Setting	Value
Physical patch extent	1.0 x 1.0 degrees
Encoder	current Perceiver encoder, depth 2
Encoder latents	1 per one-cell patch for the primary control
Encoder latent dimension	64
Learned representation width	128
Decoder base	physical-coordinate resampling plus learned 1x1 projection
Samples	32 train + 32 held out
Batch size	1

Setting	Value
Optimizer	existing Adam, no weight decay
Updates	640 (20 epochs x 32 samples)
Scheduler	none
Precision	float32/naive on L40S; retest promoted architecture on a validated flash runtime
Seed	15; add 16 only for finalists

Torch bring-up on 2026-07-21 isolated an L40S runtime failure: both selective checkpointing and checkpointing-disabled bfloat16/flash runs hit a CUDA illegal memory access on the first backward, while the otherwise matched naive-attention/float32 run completed. S1 therefore uses naive/float32 for every cell so the architecture comparison remains controlled. This is an execution runtime result, not evidence against the encoder architecture; flash/bfloat16 must be revalidated separately before a full-scale promotion run.

Screen the full {3e-4, 6e-4, 1e-3} by {additive, none} matrix on the resampling-only control. This parallelized revision avoids serial public-queue latency and reveals any learning-rate/geometry interaction directly. Freeze the best setting for subsequent structural comparisons.

Then run the encoder-geometry isolation on the promoted physical base:

Encoder geometry	Physical resampling base
Current additive position/scale	run
No post-encoder position/scale	run
Geometry sidecar	same as none at depth zero; do not duplicate

This is the primary test of the position/scale concern. Do not conclude that position embeddings are harmful from training-set MSE alone. Selection uses held-out MSE, spectra, amplitude ratios, and later processor behavior. If none wins at depth zero, sidecar must still be tested with a processor before removing geometry from the model.

Run one hybrid ocean guard only if the promoted base passes the S1 reconstruction gate but exhibits a repeatable spectral, seam, or variable-group defect. This is a deliberate evidence-based reduction from the original six-cell decoder/geometry factorial, not a permanent prohibition on the correction.

Run two capacity checks only on the best geometry/decoder pair:

- learned representation width {128, 256, 380};
- encoder latent dimension {64, 128}.

Do not sweep encoder latent count on a one-token patch beyond {1, 4}; prior runs show that manufacturing many latents for one input token adds cost without improving the identity map.

Promotion from S1 requires:

- held-out all-channel MSE no worse than the resampling-only learned-encoder control;
- no variable group more than 10% worse;
- mean high-wavenumber power ratio in [0.85, 1.15], with velocity reported separately;
- correction/base RMS below 0.5 unless the correction produces a clear held-out gain;

- no NaN, mask leakage, or longitude seam regression.

S1 primary-matrix result

The six seed-15 runs completed on Torch L40S using the controlled naive/float32 runtime described above. Metrics are from the disjoint 32-sample held-out set; `best` and `final` are reported separately because the two higher additive learning rates bounced after their minima.

Encoder geometry	LR	Best epoch	Best MSE	Final MSE	Mean high-k ratio	Mean amplitude ratio	Mean absolute bias / target std
additive	3e-4	20	0.021202	0.021202	0.8851	0.9561	0.0092
additive	6e-4	19	0.015452	0.020102	0.9264	0.9784	0.0125
additive	1e-3	19	0.013569	0.015551	0.9325	0.9814	0.0118
none	3e-4	20	0.020776	0.020776	0.8372	0.9310	0.0149
none	6e-4	20	0.013780	0.013780	0.9247	0.9745	0.0098
none	1e-3	20	0.011844	0.011844	0.9283	0.9758	0.0076

`none + 1e-3` wins every variable-group MSE against `additive + 1e-3`. Its per-variable high-wavenumber ratios are 0.929 (temperature), 0.953 (salinity), 0.894 (zonal velocity), 0.929 (meridional velocity), and 1.094 (SSH), all inside the gate. Its amplitude ratios range from 0.960 to 0.994. The selected model therefore removes additive encoder geometry at depth zero; this does not remove geometry from the eventual processor, where the sidecar remains mandatory to test.

The one-cell patch seam ratio is undefined on the one-degree grid because every edge is a patch edge and there are no within-patch edges for its denominator. Seams are evaluated on the half-degree and cross-resolution routes in S2 instead.

The promoted geometry/LR pair then produced this capacity result at the best held-out epoch:

Embedding width	Latent dimension	Best epoch	Best MSE	Final MSE	Mean high-k ratio	Mean amplitude ratio	Mean absolute bias / target std
128	64	20	0.011844	0.011844	0.9283	0.9758	0.0076
256	64	19	0.010375	0.010903	0.9378	0.9798	0.0223
380	64	20	0.009505	0.009505	0.9703	0.9956	0.0221
128	128	20	0.007937	0.007937	0.9256	0.9707	0.0135

Increasing internal encoder latent dimension is decisively better than widening the external representation. The 128/128 model wins aggregate and every variable-group MSE; its per-variable high-k ratios range from 0.889 to 0.987, inside the gate. It also keeps the processor interface at width 128. The selected S2 contract is therefore embedding width 128, latent dimension 128, geometry `none`, and learning rate `1e-3`. The zero-depth model has about 0.92M parameters; the width-380 control only rises to 0.97M there, but requiring every processor block to carry 380 channels would be the material downstream cost.

S2. Small cross-resolution ocean reconstruction

Paired same-timestamp loading is implemented in `src/samudra/datasets.py` through the default-off `target_time_mode: current` option. The identity harness uses it in `configs/samudra_multi_om4/identity_cross_1_halfdeg.yaml`, with 32 training and 32 held-out samples balanced over these four routes in every fixed sample cycle:

```
1 degree    -> 1 degree
1/2 degree  -> 1/2 degree
1 degree    -> 1/2 degree
1/2 degree  -> 1 degree
```

Use a shared encoder and decoder. Normalize each physical source with its existing dataset statistics, and compute metrics in both normalized and de-normalized units. Treat the independently regridded OM4 source as the target; also report the error of the deterministic physical resampler so the model is not credited for differences already present between dataset products.

Common settings are batch size 1, Adam, no scheduler, 1,280 optimizer updates, learning rate inherited from S1, seed 15, and balanced route sampling. Compare only:

1. winning encoder geometry plus resampling base;
2. winning encoder geometry plus hybrid;
3. current additive geometry plus hybrid as a guard against overfitting the geometry conclusion to one degree.

S2 coarse-patch control and plan revision

The first four-route control completed 40 epochs using the selected Perceiver/resampling architecture. Its resolved config was naive attention with bfloat16 (despite an inaccurate `naivefp32` run suffix); this is also evidence that the earlier L40S fault requires the flash path rather than bfloat16 alone. The final route results are:

Route	Model MSE	Mean high-k ratio	Patch-seam ratio
1 degree -> 1 degree	0.01954	1.158	undefined on one-cell patches
1 degree -> 1/2 degree	0.06027	0.508	0.898
1/2 degree -> 1 degree	0.01832	1.066	undefined on one-cell patches
1/2 degree -> 1/2 degree	0.05737	0.496	0.875

The defect follows output bandwidth, not cross-grid direction. Both half-degree outputs lose about half their high-wavenumber power, including the half-to-half route where decoder coordinates already match the target. The one-to-half learned MSE also reaches the interpolation reference while remaining spectrally coarse. This localizes the new failure upstream of decoder rendering: a single vector per one-degree physical patch cannot preserve four native half-degree cells.

This evidence revises the original three-way attention comparison. Do not run the hybrid for this defect; decoder attention cannot recover information discarded by the encoder. The existing `spatial_query_shape` path is not sufficient unchanged, either: it packs ordered query

results into channels at one coarse patch center, while a shared pointwise resampling decoder has no intra-patch position with which to select those channels. With one encoder latent its output queries are also query-blind. A future learned-query candidate must spatially unpack queries onto a finer canonical grid and use multiple or directly anchored encoder tokens.

The smaller immediate falsification test is the checked-in `CanonicalResampleEncoder`: apply an ordinarily initialized learned pointwise channel projection at every native cell, then physically resample those latent features onto the finest configured grid. The encoder remains learned and receives no identity initialization or prognostic bypass. Its latent grid is independent of the requested output, so the same state can be processed zero to N times and decoded at flexible resolutions.

The one-degree held-out screen strongly supports this route:

Encoder	External width	Held-out MSE	Mean high-k ratio	Mean amplitude ratio	Mean absolute bias / target std
Perceiver, latent dim 128	128	0.007937	0.9256	0.9707	0.0135
Canonical pointwise/ resample	128	0.002857	0.9677	0.9965	0.0020
Canonical pointwise/ resample	160	0.002197	0.9728	0.9971	0.0018

The width-160 candidate wins every variable-group MSE and is now running the matched naive/bfloat16 four-route S2 comparison. Width 128 remains the conservative processor-width fallback because it is already 64% better than the selected Perceiver control; the 160-width gain is 23% and must survive cross-resolution and processor tests before becoming permanent.

That first fixed-finest-grid comparison was stopped after epoch 10 because it exposed a second deterministic tradeoff:

Route	Epoch-10 MSE	Mean high-k ratio
1 degree -> 1 degree	0.03178	0.446
1 degree -> 1/2 degree	0.06298	0.430
1/2 degree -> 1 degree	0.03208	0.629
1/2 degree -> 1/2 degree	0.00821	0.968

It confirms the spatial diagnosis—half-to-half error falls 86% relative to the fully trained coarse-patch control and fine-scale power is restored—but it is not a shared inverse. Bilinearly upsampling one-degree latent features to the fixed half-degree grid and then decoding back to one degree is a smoothing round trip. The job was deliberately cancelled after this invariant was measured instead of spending the remaining 960 updates optimizing around it.

The next revision keeps the successful learned pointwise channel projection but leaves its spatial grid at the native input resolution. The processor is fully convolutional and can apply the same weights zero to N times on either native grid; the decoder alone performs physical-coordinate resampling to any requested output. This makes both same-grid routes exact in spatial transport, preserves target- independent encoding, and avoids both one-degree patch

compression and an upsample/downsample round trip. The checked-in candidates are `model_learned_inverse_native_projection.yaml` and `model_iterable_inverse_native_projection.yaml`.

The original deterministic half-to-one reference was invalid because destination wet cells without any source-wet interpolation support were filled with physical zero, producing normalized MSE near 39.4. The diagnostic now uses the destination channel climatology only at unsupported points and retains wet-mask-renormalized physical interpolation everywhere else. The completed model trajectory is unaffected. The corrected normalized MSE is 0.005429 for half-to-one; one-to-half remains 0.061161, and both same-grid references are numerical zero.

The native-grid run completed all 1,280 updates and passed its structural gate. Held-out same-grid MSE is 0.00358 at one degree and 0.00348 at half degree, with mean channel high-wavenumber ratios 0.992 and 1.000. One-to-half MSE is 0.06120, effectively equal to the physical interpolation reference 0.06116. Half-to-one remains 0.02934, however, or 5.4 times the corrected physical reference 0.00543. The excess survived continued optimization while both same-grid errors kept falling, so it is not evidence for more spatial encoder capacity.

That asymmetric remainder first motivated a matched normalization control before processor promotion. The encoder and decoder in this candidate are affine 1-by-1 maps and therefore commute with unmasked interpolation. With independently normalized sources, one shared map cannot be the identity on both same-grid routes and also apply two different source-to-target mean/std transforms. The checked-in `identity_cross_1_halfdeg_common_stats.yaml` keeps both native products and masks but normalizes every channel using the one-degree scalar statistics. A second diagnostic, `source_normalized_resampler_mse`, measures interpolation performed directly in each source's normalized coordinates.

The epoch-10 common-statistics result improved half-to-one MSE from 0.03459 to 0.03013 at the matched point, but did not approach its 0.00543 reference. Normalization is therefore a contributor, not a sufficient root cause. The more important non-commuting operation is channel-wise masking: the baseline resamples 160 latent channels after the encoder has mixed prognostic variables with different depth-dependent land masks, while the physical reference resamples each of 154 prognostic channels using its own wet-neighbor denominator. No shared latent mask can reproduce that operation.

The completed checkpoint-only diagnostic reaches the same conclusion without training-trajectory ambiguity. On the original independently normalized data, direct interpolation in the source-normalized basis gives MSE 0.00971 for half-to-one, versus 0.00543 when physical values are transformed into the target basis. That roughly 0.00428 normalization penalty explains only a small part of the learned model's 0.02391 excess over the physical reference.

The matched `identity_cross_1_halfdeg_common_stats_masked.yaml` control therefore projects latent features back to prognostic channels on the native source grid and only then applies coordinate resampling with the immutable source wet masks. This does not bypass the learned representation or leak targets: the encoder and decoder remain learned pointwise maps, the processor still runs zero to N times, and the masks carry geometry only. Promote this ordering if it closes the half-to-one excess without regressing same-grid reconstruction; revisit scale conditioning or learned local correction only if it does not.

The checkpoint-only decoder-order swap provides the causal result. Using the exact epoch-25 common-statistics weights, same-grid MSEs are unchanged to below $4e-9$: 0.00513708 at one

degree and 0.00564119 at half degree. One-to-half improves slightly from 0.0877352 to 0.0867701. Half-to-one falls from 0.0260986 to 0.00996908, removing 78.0% of the learned excess above the 0.00542936 physical floor. This isolates mask ordering as the primary remaining cross-grid cause. A fresh matched run is still required to measure how much of the residual 0.00454 excess is removable by optimizing with the corrected ordering from initialization.

Fresh-run bring-up agrees with the swap. At epoch 5, half-to-one MSE is 0.02121 versus 0.03737 for the common-statistics latent-resampling control at the same point; aggregate MSE is 0.03856 versus 0.04350. Same-grid MSE remains comparable (0.01681/0.01883 versus 0.01774/0.01971). At epoch 5, the masked-resampling training portion takes about 2.2 times as long (156 versus 71 seconds for 32 training samples). `identity/epoch_seconds` stops before route evaluation, so this is a training-throughput measurement rather than an evaluator timing. Quarter-degree promotion must still include independent memory and throughput gates. Subsequent runs log `identity/training_seconds`, `identity/evaluation_seconds`, and `identity/total_seconds` separately.

At epoch 10, half-to-one improves further to 0.01352, compared with 0.03013 for the matched control. Same-grid MSEs are 0.00877 and 0.00993, and aggregate MSE 0.03063 has already surpassed the control's final 0.03079 after one quarter of the optimizer budget.

At epoch 15, aggregate MSE falls to 0.02830. Half-to-one reaches 0.01121 against the 0.00543 deterministic masked-resampling floor; the one- and half-degree same-grid MSEs are 0.00634 and 0.00726. One-to-half is 0.08839 against its 0.07900 deterministic floor. The corrected ordering therefore keeps closing both learned-projection and cross-grid excess without trading away the same-grid inverse.

At epoch 20, half-to-one reaches 0.01003, effectively reproducing the checkpoint-only order swap's 0.00997 from a fresh initialization. Same-grid MSEs are 0.00509 and 0.00588, one-to-half is 0.08736, and aggregate MSE is 0.02709. This independent optimization result makes the mask-order conclusion robust to the original checkpoint's training path.

The 40-epoch run finishes at its validation best with aggregate MSE 0.02491. Final route MSEs are 0.00304 for one-to-one, 0.08498 for one-to-half, 0.00805 for half-to-one, and 0.00357 for half-to-half. Half-to-one is only 0.00262 above its 0.00543 deterministic floor, while the same-grid and half-to-one high-wavenumber ratios are 0.936, 0.938, and 0.934. Use this validation-selected checkpoint for processor-depth experiments.

The epoch-10 checkpoint also completes a zero-shot nine-route evaluation after adding the quarter-degree grid. Quarter-degree same-grid MSE is 0.01055, close to the one- and half-degree same-grid errors from the same checkpoint (0.00865 and 0.00981). One-to-quarter MSE is 0.10897 versus the deterministic interpolation floor 0.10903; half-to-quarter is 0.03747 versus 0.02959; and quarter-to-half is 0.01158 versus 0.00268. This is strong evidence that the learned native-grid inverse generalizes to an unseen resolution and that most upsampling error is the unrecoverable physical interpolation floor.

Quarter-to-one exposes a separate scale-aware transport issue. Its MSE 0.07040 beats the bilinear deterministic reference 0.07933, but aggregate high-wavenumber power is 1.35, including 1.73 for velocity. The underlying OM4 products were conservatively regridded, whereas the current decoder point-samples with bilinear interpolation even for a fourfold

restriction. A fixed-checkpoint masked regular- index area diagnostic confirms that mechanism while showing the operator must be scale aware:

Route	Bilinear floor MSE	Area floor MSE	Model high-k	Area high-k
1/2 -> 1	0.00592	0.00741	0.888	0.997
1/4 -> 1/2	0.00268	0.00399	0.874	1.010
1/4 -> 1	0.07933	0.01123	1.351	1.001

For the fourfold restriction, area averaging removes 86% of the bilinear floor and restores the spectrum. For both twofold restrictions it restores spectral power but raises MSE by 25–49%. Revise the promoted transport to use channel- masked conservative or antialiased restriction for substantially coarser outputs, retaining coordinate bilinear interpolation for prolongation. Do not promote regular-index area pooling unchanged at 2x; compare a physically weighted conservative operator and a better low-pass kernel on the fixed samples first.

The first unchunked evaluator completed in 58 minutes of model work and held about 52 GiB of GPU memory in a spot check on a 96-GiB RTX PRO 6000. The exact chunked eval-only rerun reproduces all nine route MSEs and aggregate 0.0399052 exactly. Its held-out pass takes 1,162.5 seconds (19:22.5), with peak CUDA allocation 10.30 GiB and reservation 24.43 GiB; training time is zero. The cold Slurm job takes 56:20 because its one-time SIF construction consumes about 36 minutes; the warm area-control job completes in 15:54. The bounded-memory evaluator therefore passes the engineering gate. A first cold pull also exposed 574 GiB of stale shared Apptainer cache; that recoverable cache was removed and the Slurm harness now forces OCI layers and build temporaries onto SLURM_TMPDIR, while retaining the keyed final SIF in shared scratch. The clean rerun is Slurm job 14574139 (W&B pod127bl); the area diagnostic is job 14579543 (W&B rqlh7bke).

After selecting among them, add quarter degree first with identity_eval_only: true, finetune: true, epochs: 1, and the selected checkpoint. The completed zero-shot result is finite and geometrically sensible except for the expected large-ratio bilinear restriction aliasing above. Validate the bounded-memory and scale-aware restriction changes, then repeat the balanced screen with all nine input/output pairs among 1, 1/2, and 1/4 degree for 2,880 updates. Diagnose any remaining coordinate or capacity failure before training on quarter-degree targets.

The earlier physical-patch and subcell-query ranges are no longer a mandatory sweep. Retain them as fallback hypotheses only if the native-grid candidate fails or its measured quarter-degree processor cost is unacceptable. In that event, start from physical patch extent {1x1, 2x2 degrees}, encoder latent count {16, 32}, and an explicitly spatially unpacked coordinate-query design; revise those values using the failure measurement rather than treating them as fixed.

S3. Processor-depth contract

After D4, pretrain the selected encoder/decoder at k=0, then introduce the processor without freezing either head.

Use finetune: true, the zero-depth checkpoint path, and the explicit missing-key allowlist above. Keep the encoder, decoder, embedding width, and geometry mode identical across the checkpoint boundary; only the processor and its optional geometry conditioner may be newly initialized.

The first diagnostic uses depths `{0,1,2,4}` and logs a decode at every depth. Keep the latent width fixed at the selected canonical width. Use two loss-weight checks:

```
lambda_0 in {0.05, 0.2}
```

plus `lambda_0=0` as the forecast-only control. Do not sweep processor architecture simultaneously. The zero-depth held-out reconstruction MSE may worsen by at most 20% from its pretrained value. If it worsens more, first test alternating reconstruction and forecast batches before increasing `lambda_0`.

The zero-depth term is now explicitly a source-grid MSE. It builds a reconstruction context from `input_resolution_cpu` and the immutable `input_mask`, decodes before any processor call, and never uses the batch's possibly different target grid. This makes the inverse constraint valid on mixed-resolution forecast batches without a prognostic bypass or target leakage. The exact staged config is `identity_cross_1_halfdeg_common_stats_masked_processor.yaml`; it initializes only `processor.*` and `processor_geometry.*` outside the selected zero-depth checkpoint.

An initial one-GPU bring-up completed successfully as Slurm job `14563629` (W&B run `joo4bptt`). It trained the `lambda_0=0.05` candidate for five epochs and 160 optimizer updates on 32 samples per epoch, then decoded the same held-out latent state at all requested depths:

processor depth	held-out MSE
0	0.021896
1	0.022411
2	0.030988
4	0.107452

Depth one is only 2.4% worse than depth zero, while depths two and four degrade substantially on every resolution route. This is evidence that the interface is legal and trainable, but that repeated-processor stability is not obtained merely by training a single call. It does not yet distinguish inadequate exposure to positive depths from intrinsically unstable processor dynamics. Exact-window baseline job `14567329` evaluates the untouched S2 checkpoint at `0.024567` MSE. The smoke's depth-zero MSE is therefore 10.9% better, and its depth-one MSE remains 8.8% better, so the 20% reconstruction-preservation gate passes. Depth two is 41.5% worse than the smoke's depth zero and depth four is 391% worse. The matched longer `lambda_0=0` and `lambda_0=0.05` jobs remain the decision runs; this smoke result is not a substitute for them.

The matched decision runs are Slurm jobs `14581788` (`lambda_0=0`) and `14581789` (`lambda_0=0.05`). Each uses two RTX PRO 6000 GPUs and accumulation four, preserving effective global batch eight, 256 samples and 32 optimizer updates per epoch, and 1,280 total training samples. They load the same validation-selected S2 checkpoint and initialize only the 65 allowlisted processor/geometry tensors. Earlier four- L40S duplicates `14562662` and `14562663` were canceled while still pending after both RTX jobs passed startup validation; they consumed zero runtime.

Both matched RTX runs complete successfully (W&B `s8rig9hd` and `u1ki1obr`):

<code>lambda_0</code>	depth 0	depth 1	depth 2	depth 4
0	0.024406	0.022151	0.033715	0.135192
0.05	0.021861	0.022284	0.030805	0.113853

The untouched exact-window checkpoint is 0.024567. Thus `lambda_0=0` merely preserves it (0.7% better), while `lambda_0=0.05` improves depth zero by 11.0%. The regularized arm is only 0.6% worse at depth one and is 8.6% and 15.8% better at depths two and four. Promote `lambda_0=0.05`. However, its depth-four MSE is still 5.2 times depth zero and is slightly worse than the short smoke's 0.10745. More single-depth exposure therefore does not make the shared map stable for arbitrary iteration. Before proxy forecasting, train positive depths sampled from {1, 2, 4} toward the same refinement target, retain the source-grid depth-zero term, and require improvement at depths two/four without regressing depth zero/one.

That causal test is complete as Slurm job 14590678 / W&B 9yart3ax. It uses the same seed, checkpoint, 1,280 samples, 160 optimizer updates, effective batch eight, and `lambda_0=0.05`; the only training change is a deterministic balanced cycle over positive depths {1,2,4}:

positive-depth training	depth 0	depth 1	depth 2	depth 4
fixed depth 1	0.021861	0.022284	0.030805	0.113853
cycle {1,2,4}	0.022202	0.021475	0.021805	0.024028

Balanced depth exposure improves depths one, two, and four by 3.6%, 29.2%, and 78.9%, respectively, while depth zero changes by only +1.6% and remains 9.6% better than the untouched exact-window checkpoint. Depth four is now only 8.2% worse than depth zero rather than 5.2 times worse. Every depth-four route improves: the four route MSEs contract from 0.08589--0.14571 to 0.00594--0.07354, with the remaining maximum on the known one-degree-to-half-degree interpolation route. This supports inadequate positive-depth exposure as the repeated-iteration root cause. Do not add a global processor residual solely to fix this failure; advance the multi-depth checkpoint to the forecast proxy and retain residualization only as a fallback if forecasting reintroduces iteration drift.

These completed runs establish only a same-target refinement control. They do not establish the physical-time contract above and must not be used to promote a latent-autoregressive model.

Forecast and full-scale validation

Only one structural candidate and one conservative fallback should enter this stage. The fallback is the learned Perceiver encoder plus physical resampling projection without the attention correction.

V0. Calibrated one-degree proxy

The results recorded below are historical. Their depth cycling used the same `t+1` target at depths one, two, and four. One-step values remain representation evidence; the multi-depth values are not physical rollout evidence.

Use the existing stratified 512-window, one-step proxy and matched v2 control:

- 12 epochs and 192 optimizer updates;
- plain normalized MSE;
- effective global batch 32;
- Adam with learning rate $6e-4$ unless S1 shows a decisive alternative;
- optimizer-update schedule targeting 6,160 updates;
- seeds 15 and 16, adding 17 only for the winner;
- no residual prediction, prognostic bypass, spectral loss, or dynamic loss.

Run:

1. resampling-only fallback;
2. hybrid candidate;
3. hybrid candidate with `lambda_0=0.05` zero-depth reconstruction regularization;
4. matched v2 reference if its existing result is not directly comparable after data or training-loop changes.

The selected inverse candidate is instantiated by

`train_ldeg_iterable_inverse_masked_mse_stratified_updates_proxy.yaml`. Its defaults use batch 2, eight ranks, and two accumulation steps for effective global batch 32 and 192 optimizer updates. Set `model.zero_depth_reconstruction_weight=0` for the forecast-only control rather than changing any other architecture or schedule.

The current references are proxy MSE `0.051655` for the Perceiver-encoder/resampling decoder, `0.041278` for direct one-cell, and `0.042390` for matched v2. Promote a candidate only if its two-seed mean is at most `0.05`, beats persistence, retains the S1 zero-depth contract, and has no material spectral or variable-group regression. Prefer the simpler fallback if the hybrid improves aggregate MSE by less than 5%.

The first two-seed forecast proxy is complete from the multi-depth S3 checkpoint. It used 512 stratified samples per epoch, 12 epochs, 192 optimizer updates, and effective global batch 32. Validation selected the terminal epoch in all four runs:

source-grid zero-depth weight	seed 15	seed 16	mean
0	0.0315724	0.0315876	0.0315800
0.05	0.0316064	0.0315465	0.0315765
0.2	0.0315684	0.0316211	0.0315947

All three weights are forecast-equivalent within 0.06%; all beat the matched v2 proxy by about 25.5%, the old Perceiver/resampling result by 38.8–38.9%, and the direct one-cell proxy by about 23.5%. Regularization improves velocity high-wavenumber power without a material variable-group regression; weight 0.2 raises the two-seed zonal ratio to about `0.733`, versus about `0.718` at 0.05. The W&B runs are `bznwj3is`, `h73bbty2`, `4zaqxkgw`, `gr0pvlvr`, `fbmv4ccl`, and `4mhuysks`.

Post-forecast reconstruction and depth audits expose the remaining failure. On the same held-out cross-resolution identity window, the two-seed depth-zero mean is `0.0374764` for weight zero and `0.0281127` for weight 0.05. Relative to the pre-forecast multi-depth checkpoint's `0.0222023`, those regress by 68.8% and 26.6%. Thus 0.05 is useful but narrowly misses the

provisional 25% preservation gate. On a separate true $t+1$ target, decoding the same input after successive processor calls gives:

zero-depth weight	depth 0	depth 1	depth 2	depth 4
0	0.0137334	0.0478494	0.0968079	0.291519
0.05	0.00631293	0.0476432	0.0880912	0.219623

Depth zero is a persistence-like inverse control on this future-target audit; depth one is the trained one-step forecast. Applying the processor two or four times toward that same refinement target deteriorates sharply. This reproduces the S3 exposure failure under forecast training and rules out treating successful multi-depth autoencoding as a permanent property of the checkpoint. Regular training now has a default-off `train_processor_depths` option that cycles positive depths continuously and identically on every rank, restores the configured depth for validation, and logs the selected depth. The iterable-inverse proxy and full configs select `[1, 2, 4]`. The next matched proxy must use this exposure before a full-scale launch.

The weight-0.2 fixed-depth inverse audit passes: its two-seed depth-zero mean is `0.0241583`, an 8.8% regression from the pre-forecast `0.0222023`, versus 26.6% for weight 0.05. The matched multi-depth rerun therefore used weight 0.2 and the same two seeds:

forecast training depths	seed 15	seed 16	mean
fixed <code>{1}</code>	0.0315684	0.0316211	0.0315947
cycled <code>{1,2,4}</code>	0.0360968	0.0361911	0.0361440

At the deliberately fixed 192-update budget, multi-depth exposure costs 14.4% aggregate MSE but still beats matched v2 by 14.7%, direct one-cell by 12.4%, and the old Perceiver/resampling proxy by 30.0%. Its two-seed $t+1$ depth audit is:

depth	0	1	2	4
mean MSE	0.00371387	0.0413635	0.0455987	0.0485074

Depth four is only 17.3% worse than depth one, compared with roughly 4.6 times worse under fixed-depth training. The current-state inverse mean is `0.0246888`, only 11.2% worse than the pre-forecast checkpoint. Those facts validate the same-target refinement control only. They do not pass the corrected physical V0 gate and do not promote inverse weight `0.2`. The historical forecast W&B runs are `9i47kib4` and `t7855wfw`; their depth audits are `dp1y2chf` and `3yluahc9`; their inverse audits are `142he1n3` and `hufdi48y`.

The corrected V0 sequence is:

1. Retrain the selected state-only encoder and decoder from scratch on zero-depth autoencoding, first one degree through the standard trainer with `train_1deg_state_only_autoencoder_proxy.yaml` (the equivalent identity-only diagnostic is `identity_1deg_state_only_native_masked_projection.yaml`), then matched one/half-degree routes with `train_cross_1_halfdeg_state_only_autoencoder_proxy.yaml` (with `identity_cross_1_halfdeg_common_stats_masked.yaml` retained for detailed route diagnostics).
2. Confirm same-grid, cross-grid, and unseen quarter-degree reconstruction before introducing a processor. Compare against the prior joint-encoder numbers, but do not load that checkpoint.

3. Initialize a fresh boundary encoder and processor from the new inverse. Train physical leads {1,2,4} with four-step data windows, aligned forcing at every call, and true targets {t+1,t+2,t+4}.
4. Run seeds 15 and 16 at the existing 512-window/12-epoch proxy scale. Sweep zero-depth weight {0,0.05,0.2}; use learning rate 6e-4, effective batch 32, and 192 updates as starting values subject to the selection-logic revision rule.
5. Audit decoded MSE at each true lead, zero-depth inverse retention, forcing ablations/shuffles, spectra, and latent chunk continuity. Promote only after the predicted lead beats persistence and the processor is demonstrably sensitive to the correctly aligned boundary state.

Corrected V0 execution is now underway on exact, checksum-verified code overlays. The first clean-break result is decisive: the state-only one-degree inverse (14616192, W&B 4xbqyev0, code b3f7d823) reaches held-out MSE 0.000648411 after 40 epochs. The prior joint prognostic-plus-boundary inverse reached about 0.00304. The exact fixed-window audit 14619746 confirms the result at 0.000654260, a 78.5% reduction with high-wavenumber power ratio 0.988 and standard-deviation ratio 0.999.

The fresh one/half-degree mix run (14618325, W&B 27t8occ3, code f4a57cb8) completes at aggregate MSE 0.0210660, split as 0.00365078 on one-degree outputs and 0.0384813 on half-degree outputs. The corresponding deterministic persistence / coordinate-resampling baselines are 0.00259272 and 0.0372132. Its exact route audit 14619373 is more informative: relative to the earlier joint encoder, aggregate MSE improves from 0.02491 to 0.0218603; one-to-one from 0.00304 to 0.00108228; one-to-half from 0.08498 to 0.0789610; half-to-one from 0.00805 to 0.00617820; and half-to-half from 0.00357 to 0.00121959. Every route improves even though the corrected checkpoint trained for 10 rather than 40 epochs. One-to-half is fractionally better than its 0.0789996 deterministic floor; half-to-one has 0.0007488 excess over its 0.00542936 floor, and the same/fine-to-coarse high-wavenumber ratios are 0.979--0.985. This is direct evidence that removing boundary channels from the learned state is beneficial, not merely harmless.

The nine-route audit 14619374 then applies that one/half-trained checkpoint to the unseen quarter-degree product. Quarter same-grid MSE is 0.00136714, versus 0.01055 for the earlier joint encoder. One-to-quarter is 0.0974551 versus its 0.109031 deterministic floor; half-to-quarter is 0.0260758 versus 0.0295858; and quarter-to-half is 0.00322509 versus 0.00268161. The same-grid quarter high-wavenumber ratio is 0.958, and quarter-to-half is 0.986. Thus flexible output resolution survives the clean state separation and improves substantially. Quarter-to-one remains the known scale-aware restriction exception: MSE 0.0698314 beats bilinear resampling's 0.0793258, but its aggregate high-wavenumber ratio is 1.502. This supports an antialiased/conservative deterministic restriction path for large downsampling factors; it does not revive an unstructured attention decoder or justify adding the zero-initialized attention residual everywhere.

A two-epoch physical-time bring-up (14619138, W&B o6gi7wvi, code 455be788) loaded the state-only inverse, initialized only the boundary encoder and processor, and completed the encode-once rollout without decode/re-encode. True-lead validation improved from {0.101246, 0.165935, 0.277784} after epoch 1 to {0.0811572, 0.132833, 0.198276} after epoch 2 for leads {1,2,4}. This is a functionality and optimization smoke, not a promotion result. The definitive two-seed, three-weight proxy jobs use code ed7e7cb9 and additionally log a lead-

matched persistence baseline, masked zero-depth inverse retention, and zeroed, batch-shuffled, and time-reversed boundary controls.

The completed two-seed weight sweep isolates representation-head drift rather than an inverse-loss tradeoff:

zero-depth weight	zero-depth reconstruction	lead 1	lead 2	lead 4
0	0.0129752	0.0430257	0.0784274	0.116421
0.05	0.00938407	0.0429238	0.0783425	0.116196
0.2	0.00510098	0.0430606	0.0784412	0.116427

All three weights are forecast-equivalent, and stronger inverse supervision is Pareto-improving, but even weight 0.2 leaves reconstruction 7.9 times worse than the frozen checkpoint's 0.000647529. Direct checkpoint comparison supplies the mechanism: after epoch nine, the unregularized encoder and decoder move 18.7% and 14.6% in relative parameter norm and their composed pointwise map moves 27.3%; weight 0.05 only reduces the composed-map drift to 25.9%. Forecast loss constrains $D(P(E(x)))$, not $D(E(x))$, so co-adaptation can destroy the inverse without hurting the supervised objective.

Commit 2483dd00 adds a fail-loud prefix freeze. Job 14623982 freezes the already learned encoder.* and decoder.* while fitting only the processor, boundary encoder, and processor geometry. This does not make the encoder trivial: it preserves the state-only representation learned above. Its final seed-15 leads are {0.04445, 0.07987, 0.11840} and zero-depth reconstruction remains exactly 0.000647529 at every epoch. Freezing is therefore sufficient to preserve the inverse, but the replacement-form processor is less well conditioned.

Commit 74b6f1de adds a per-channel ReZero transition $z[m+1] = z[m] + \alpha * F(z[m], b[m])$, with alpha initialized to zero. This initializes every processor call as latent persistence, remains iterable from zero to N calls, and is distinct from the rejected decoder-attention residual. Combined with the frozen inverse, the two-seed mean is {0.04019, 0.06704, 0.09109} for leads {1,2,4} while retaining the exact 0.000647529 reconstruction in both seeds. Relative to the two-seed unfrozen baseline, this improves the three leads by 6.6%, 14.5%, and 21.8% despite giving up all head adaptation. The seed-15-only frozen replacement control is {0.04445, 0.07987, 0.11840}; the residual form improves those leads by 9.1%, 15.6%, and 22.6% on the matched seed. At the spatial audit, its temperature, salinity, zonal-velocity, meridional-velocity, and SSH high-wavenumber ratios are {0.956, 0.968, 0.786, 0.743, 0.990}.

The forcing controls establish physical use of the separate boundary path. At the two-seed endpoint, zeroing boundary input worsens leads {1,2,4} by {4.9%, 14.9%, 30.7%}, and reversing its time order worsens lead four by 37.7%. The two seeds differ by only about 1.3% at each lead, far inside the gap to the alternatives. The frozen ReZero design is therefore promoted to full one-degree validation; its checked-in selection is commit d011fce5. Runtime-resume fixes through a70410f3 are used by Torch job 14629647.

Three failed setup jobs contribute no model evidence: 14616135 invoked an identity config through the training entry point, 14616181 duplicated the data subdirectory in DATA_ROOT, and 14616295 exposed the now-fixed cross-grid persistence shape bug after training but before completing epoch-one validation. The route-audit harness failure 14619195 likewise performed

no evaluation; the harness now explicitly supports `samudra.identity` and its replacements are pinned to the completed cross-resolution checkpoint.

V1. Full-data one-degree run at v2-like scale

After corrected V0 promotion, train its winner on the 1975–2013 one-degree training interval and 2013–2014 validation interval:

- reference budget: 70 epochs at effective global batch 32 for 6,230 realized optimizer updates;
- actual fragmentation-adjusted budget: 65 epochs at effective batch 30 for 6,110 updates;
- Adam, initial learning rate $6e-4$, cosine schedule in optimizer-update units;
- normalized MSE and absolute-field prediction at true physical leads $\{1,2,4\}$, selecting one lead per training batch;
- seed 15 first; run seed 16 only if seed 15 passes the gate;
- evaluate the validation-selected checkpoint and terminal checkpoint;
- retain zero-depth reconstruction evaluation at every image-validation epoch.

The corresponding checked-in full-data config is

`train_1deg_iterable_inverse_masked_mse_updates.yaml`; it uses the same learned heads, processor sidecar geometry, optimizer-update schedule, and effective global batch as the proxy.

Seed 15 runs from exact code/image commit `f5366fdd89dfb82c1e6f42a9d00b17939c440a41`. Cluster fragmentation left no node with eight available RTX6000s, so it uses two GPUs with gradient accumulation eight to preserve effective global batch 32 and the exact 6,230-update schedule. Initial job `14605300` made no optimizer step: both ranks timed out in the first NCCL barrier on `gr102`. The first scratch-backed replacement, job `14605887`, also made no optimizer step because a silently corrupted SIF copy failed to mount; its digest differed despite matching size and a successful copy exit. The SIF was rewritten with checksummed `rsync`, verified against SHA-256 `0667e69d7079823cb2ffda2d15784c282ff176c2494cc18fea3d3a24b4123729`, and passed `aptainer inspect`. Job `14607299` was the exact replacement with Torch's documented `NCCL_P2P_DISABLE=1` workaround and scratch-backed logs, caches, and verified SIF. It completed about 88 updates (one epoch) before being cancelled immediately after the physical-time semantics were clarified. It is a mis-specified run and supplies no model-selection evidence. Jobs `14605300` and `14605887` made zero updates.

The original eight-GPU submission `14626058` made no optimizer step. Node fragmentation gave it an estimated start two days later, so the selection-logic revision rule replaces it with Torch job `14629647` from exact code overlay `a70410f3`. The replacement uses the three immediately available RTX6000s, per-rank batch 2, accumulation 5, and 65 epochs: effective batch 30 and about 6,110 updates, 1.9% below the requested 6,230-update budget. It resumes the same frozen state-only inverse checkpoint and is the requested v2-scale validation: full one-degree data, roughly the existing v2/Samudra parameter and update scale, and the same primary metrics. Compare against the quoted v2 full-data MSE `0.023600` and the direct one-cell result `0.015976`. Promotion requires all-channel MSE at most `0.025`, no variable group more than 25% worse than v2, and zero-depth reconstruction no more than 25% worse than its pre-forecast checkpoint.

The first three-GPU launch `14629149` verified world size, effective batch, head freeze, ReZero selection, checkpoint provenance, and six optimizer updates, then was cancelled. Its measured

startup throughput projected beyond Torch's enforced 24-hour limit, and the scheduler rejected a longer allocation. The first requeueable replacement, 14629179, exposed an initialization guard that rejected `finetune + preemptible` before optimizer construction and made no update. Commit 4c7365f3 revises the clean-break contract: an initial preemptible run may load the inverse by finetuning, while a requeue that finds `ckpt.pt` disables finetuning and restores full model, optimizer, scheduler, counter, EMA, and W&B state. Job 14629304 completed epoch one and produced the first full-data metrics, then a deliberate requeue exposed a read-only provenance-manifest overwrite in the Slurm harness. Commit d64ca8f4 makes restarts compare the pinned manifest rather than overwrite it; a70410f3 also normalizes the unused finetune allowlist on an explicit full-state resume. Job 14629647 loads the epoch-one `ckpt.pt` with `Start Epoch: 2`, the saved optimizer learning rate, counters, EMA, and W&B ID, and continues with epoch checkpoints, Slurm requeue, and a five-minute pre-timeout signal. The six-update bring-up is startup evidence only and is not included in model metrics.

The full-data validation curve descends monotonically through the useful training range. Lead-one MSE falls from 0.05116 at epoch one to 0.0250324 at epoch 17, which misses the gate by only $3.24e-5$, then passes at epoch 18 with a physical-lead vector of {0.0246967, 0.0413005, 0.0551152} for {1,2,4}. These beat lead-matched persistence by {70.5%, 72.7%, 75.4%}. Zero-depth reconstruction remains exactly 0.000647529 at every point. At epoch 11, temperature, salinity, zonal velocity, meridional velocity, and SSH high-wavenumber ratios are {0.967, 0.976, 0.788, 0.793, 0.990}. Every lead-one variable group is better than its quoted v2 value. V1 therefore passes its promotion gate without rounding the epoch-17 near miss.

Job 14629647 completes all 65 epochs in 8 hours 28 minutes. Epoch 58 is the validation-selected checkpoint with {1,2,4} lead MSE {0.0205191, 0.0343797, 0.0469026}. The terminal checkpoint is close at {0.0205354, 0.0344690, 0.0472080}, while zero-depth reconstruction remains exactly 0.000647529. The final image-validation audit at epoch 61 measures high-wavenumber ratios {0.968, 0.983, 0.811, 0.839, 0.991} for temperature, salinity, zonal velocity, meridional velocity, and SSH. The slight late plateau supports checkpoint selection but does not change the architecture conclusion.

Profile before launch. Existing evidence suggests one RTX6000 is sufficient for proxy screening; the completed full direct one-cell run used two GPUs, peaked near 17.3 GiB per rank, and took about 8 hours 20 minutes. The hybrid must demonstrate bounded-neighborhood memory scaling in a one-epoch smoke before receiving the full budget.

Matched one-GPU throughput profiles exposed a runtime boundary rather than a model capacity boundary. Per-rank batches 16 (14624768), 8 (14625028), and 4 (14625151) all fail in the first backward with a CUDA illegal-address error, despite reported total-size estimates falling from 15.25 to 4.03 GiB. No optimizer step occurs in any of those runs. Batch 2 is stable across every scientific proxy run. The full validation therefore retains per-rank batch 2; diagnosing the larger-microbatch CUDA kernel is orthogonal to the architecture decision.

V2. Full multi-resolution validation

Do not begin with a three-resolution full-budget run. Use two gates:

1. Train 1-degree plus 1/2-degree on the full time interval with balanced resolution sampling and a total compute-matched budget of 6,230 optimizer updates.

2. If both resolutions pass their same-grid and cross-grid gates, add quarter degree and repeat at 6,230 total updates.

Use one resolution pair per optimizer update and report sample counts by route. Set microbatch size independently by resolution, using gradient accumulation to preserve effective global batch 32; begin with batch 2 at one degree and batch 1 at half and quarter degree. Do not interpret the compute-matched run as converged if each resolution receives fewer updates. For the final winner only, run an exposure-matched budget of 6,230 updates per resolution (12,460 for two resolutions or 18,690 for three).

Evaluate every same-resolution route and every cross-resolution route at fixed held-out timestamps. Promotion requires:

- no same-resolution all-channel MSE more than 25% worse than the corresponding single-resolution model;
- no cross-resolution route worse than the deterministic coordinate-resampling reference by more than 10% unless it materially improves high-wavenumber fidelity;
- high-wavenumber ratios between 0.8 and 1.2 for temperature, salinity, and SSH, with velocity ratios reported and required to improve over the resampling-only fallback;
- no longitude seam, latitude-edge, mask, or grid-shift artifact;
- stable zero-depth reconstruction throughout processor training.

The checked-in preparatory proxy config

`train_cross_1_halfdeg_iterable_inverse_masked_mse_stratified_updates_proxy.yaml` uses 128 stratified samples for each of the four one/half-degree routes, 12 epochs, and 192 optimizer updates at an eight-GPU effective batch of 32. It freezes the selected state-only encoder/decoder inverse and trains only the ReZero processor, processor geometry, and separate boundary encoder. It is a constant-learning-rate screen against the full 6,160-update scheduler horizon, not a convergence claim. The proxy uses the half-degree-tested per-rank microbatch of one and retains zero and time-reverse forcing controls; the batch-shuffle control remains covered by V1. Commits `92b6347f` and `ebad49e4` report validation losses by exact input/output grid route for the aggregate forecast, every physical lead, persistence, zero-depth reconstruction, and each forcing ablation. On image-validation epochs they also isolate route-level spatial spectra, high-wavenumber retention, seam, edge, and amplitude diagnostics. Existing overall and destination-grid metrics remain available. This closes a measurement ambiguity that would otherwise pool one-to-one and half-to-one forecasts under the same output resolution.

After V1 passed at epoch 18, Torch job `14635707` was submitted from immutable code layer `0fc36dfd` with an `afterok:14629647` dependency. It requests two RTX6000s, batch one, and accumulation 16, preserving effective batch 32 and the exact 192-update proxy budget. The dependency prevents resource preemption and starts V2 only after V1 completes successfully. The dependency releases after V1 exits successfully, and V2 starts on the same node. Bring-up verifies world size two, code commit `0fc36dfd`, code-layer SHA-256 `9d56424b706ee0eb45d5aaed60b6b7ccc6ad538eddd568c13cf02fe06a28d044`, the selected cross-grid inverse checkpoint, four frozen encoder/decoder parameter tensors, batch one, accumulation 16, effective global batch 32, and first-epoch training plus validation. Its W&B run is `i11mmskt`.

Job 14635707 completes all 12 epochs and exactly 192 optimizer updates in 1 hour 34 minutes. Epoch 12 is validation-selected with aggregate {1,2,4} lead MSE {0.0629945, 0.0945314, 0.123373}, beating lead-matched persistence by {41.4%, 46.8%, 51.2%}. Route-level results are:

Route	Lead 1 MSE / persistence reduction	Lead 2	Lead 4
1 -> 1	0.045609 / 45.5%	0.076865 / 49.1%	0.103631 / 53.8%
1 -> 1/2	0.108017 / 22.3%	0.134762 / 29.3%	0.163326 / 34.5%
1/2 -> 1	0.041871 / 52.7%	0.069580 / 55.4%	0.095643 / 58.3%
1/2 -> 1/2	0.056481 / 52.3%	0.096918 / 54.3%	0.130894 / 57.4%

The frozen inverse remains 0.00111017 on one degree and 0.00124296 on half degree. At the epoch-seven spatial audit, tracer and SSH high-wavenumber ratios are 0.922--1.008; same-grid velocity ratios are 0.823--0.866; and all defined route seam ratios are 0.843--1.028. One-to-half velocity ratios are only 0.278/0.125 because deterministic prolongation cannot create absent fine-grid velocity modes. This is an information-resolution limit, not evidence of query blindness or mask-order failure.

The proxy is deliberately too short to compare same-grid forecast MSE with the converged one-degree run. Its purpose is satisfied: all routes optimize, every route beats its own persistence baseline, the inverse stays fixed, spatial diagnostics expose rather than pool coarse-to-fine limitations, and boundary controls remain causal at lead four. The selection logic therefore promotes the architecture to the full one/half-degree budget. The checked-in config `train_cross_1_halfdeg_iterable_inverse_masked_mse_updates.yaml` uses the full training interval, 17 epochs, effective global batch 30 on six GPUs, and an expected 6,392 optimizer updates, within 3% of the one-degree reference budget.

The full run is Torch job 14666431, W&B run 2ruaaimh, launched from immutable code layer 122ef519 on six RTX6000s. Bring-up verifies the frozen inverse, separate boundary encoder, effective global batch 30, and 376 optimizer updates per epoch. Epoch one already beats route-aggregated lead-matched persistence by {44.9%, 49.6%, 53.7%} at leads {1,2,4} while holding zero-depth reconstruction at 0.00117656.

The job completes all 17 epochs and exactly 6,392 optimizer updates in 10 hours 30 minutes. Epoch 17 is both validation-selected and terminal, with aggregate lead MSE {0.0398245, 0.0540801, 0.0659528}, respectively {62.9%, 69.5%, 73.9%} below lead-matched persistence. Every one/half-degree route beats persistence at every lead. Zeroing aligned boundary input raises the aggregate leads by {23.4%, 61.1%, 124.7%}, while reversing boundary order raises them by {9.8%, 19.2%, 93.7%}. Same-grid reconstruction remains exactly 0.00111017 and 0.00124296.

The immutable launch config logs expensive spatial diagnostics at epochs 1 and 18, but the run ends at epoch 17. Restarting a healthy full-budget run solely to change that cadence would discard useful compute. The branch therefore adds a general `validation_only` training mode and `validate_cross_1_halfdeg_iterable_inverse_masked_mse.yaml`. It loads an explicit checkpoint into a separate W&B run, executes one full validation pass with route and spatial diagnostics, performs no optimizer step, and writes no checkpoint. The checked-in training config now uses a 16-epoch interval so future 17-epoch runs naturally audit epochs 1 and 17. Job 14687084 audits the validation-selected checkpoint from immutable code layer 10005b33 and completes in

5 minutes 48 seconds without an optimizer or checkpoint write. Its scalar metrics reproduce the source run within $6e-7$; W&B run `ke7l2j7a` records route-level endpoint spectra and seams. Tracer/SSH high-wavenumber ratios are $0.938-1.025$ on all routes. Fine-to-coarse velocity is $0.884/0.872$; half-degree same-grid velocity is $0.790/0.751$; and coarse-to-fine velocity remains information-limited at $0.288/0.149$.

The review boundary is the completed one/half-degree run, its selected-checkpoint endpoint audit, and the final root-cause report. A proposed scale-aware spherical conservative restriction is implemented and unit-tested at commit `10005b33`, but the nine-route quarter-degree audit was cancelled before it evaluated a checkpoint or made an optimizer update. Quarter-degree validation and training are deliberately deferred until after review; the prototype is not a selected result. The full multi-resolution processor also misses the provisional one-degree degradation gate (28.7% versus 25%), so the report recommends reviewing per-route exposure or processor capacity before quarter-degree training while retaining the selected frozen inverse architecture.

Selection logic

The plan is intentionally falsifiable and open to revision. New measurements, implementation constraints, or better hypotheses may change the experiment order, candidate set, sweep ranges, budgets, and promotion gates. The numerical thresholds and exact requirements below are initial decision aids, not immutable specifications. Any departure should be recorded with the evidence and reasoning that motivated it so the plan evolves deliberately rather than being followed mechanically.

- If removing additive encoder geometry improves $k=0$ but hurts processor runs, retain geometry as a sidecar rather than restoring content addition automatically.
- If additive geometry is best on held-out reconstruction and forecasting, keep it; the concern is then resolved empirically.
- If the hybrid does not beat the physical resampling base by at least 5% on the proxy or by a clear spectral margin, ship the simpler base.
- If raw attention values are no better than normalized values, LayerNorm is not the limiting residual-path concern; it still remains absent from the stable base.
- If cross-resolution error is dominated by encoder patch compression, increase learned subcell representation capacity before changing decoder attention.
- If zero-depth reconstruction collapses during processor training, adjust the training mixture or processor interface before making the encoder more rigid.
- If the full one-degree candidate cannot reach MSE 0.025 , do not spend the three-resolution exposure-matched budget.

Expected artifacts

Each stage should produce:

- checked-in configs or a generated sweep manifest with one immutable row per run;
- resolved configuration, commit/container identity, random seed, sample indices, and optimizer-update count;
- train and held-out reconstruction tables by variable and depth;
- forecast, persistence, and deterministic-resampling comparisons;

- spectra, high-wavenumber ratios, seam diagnostics, amplitude ratios, and maps;
- peak GPU/host memory and throughput;
- a short decision record saying which arm advances and which hypothesis was falsified.

The final report updates `docs/experiments/perceiver_decoder_root_cause.md` without replacing its decoder diagnosis. The completed evidence selects the learned native-grid projection plus projection-before-channel-masked coordinate resampling, with processor geometry in a sidecar and no attention correction. The full one/half-degree run and its selected-checkpoint endpoint audit confirm that the frozen inverse survives true latent autoregression while exposing processor/exposure limitations that stop promotion before quarter-degree training.